

Exercise: brittle fracture and the Fracture Analysis Diagram

A ferritic steel for a pressure vessel. Its Fracture Analysis Diagram is below, with all temperatures counted from the nil-ductility temperature NDT measured in the drop-weight test.

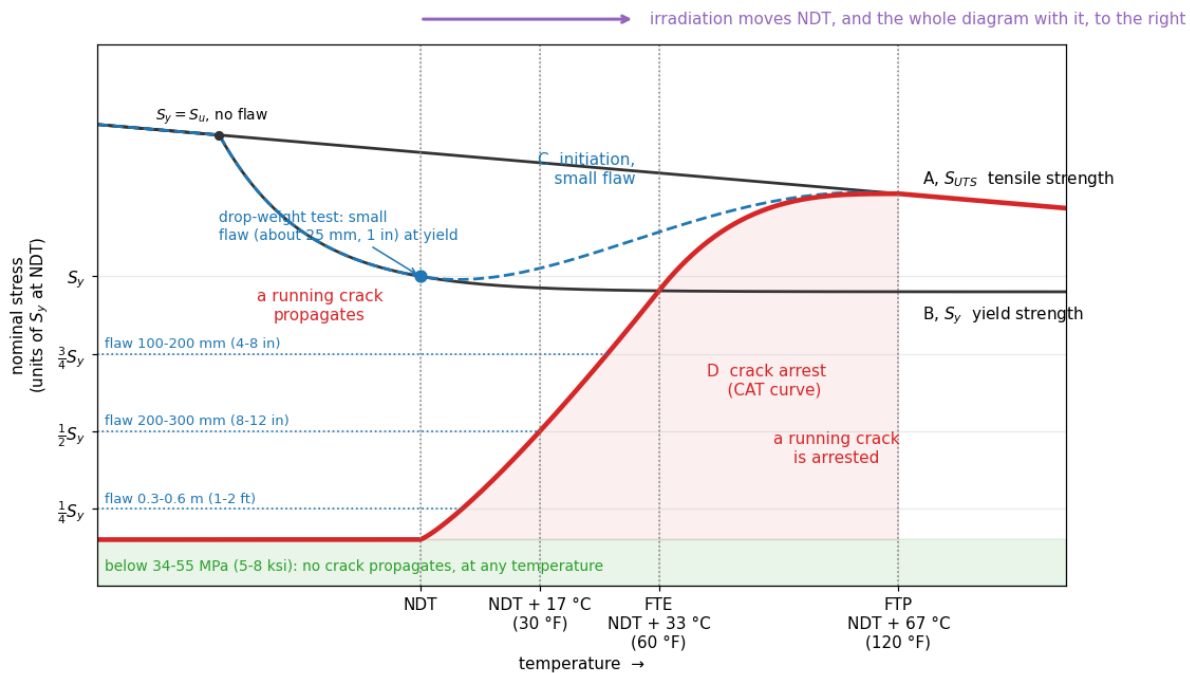


Figure 1: Fracture Analysis Diagram

data	value
yield strength	$S_y = 300$ MPa
tensile strength	$S_u = 480$ MPa
lower fracture propagation stress	34-55 MPa (5-8 ksi)
crack-arrest curve at $\frac{1}{2}S_y$	NDT + 17 °C (30 °F)
FTE, crack-arrest curve at S_y	NDT + 33 °C (60 °F)
FTP, crack-arrest curve at S_u	NDT + 67 °C (120 °F)

Temperature differences in °F convert with the factor 5/9 alone; absolute temperatures with $(T_F - 32) \cdot 5/9$.

Questions

- A crack is already running through the steel. Does it keep running or is it arrested in each case?
 - NDT - 10 °C, 30 MPa; (b) NDT - 10 °C, 150 MPa; (c) NDT + 45 °C, 250 MPa;
 - NDT + 20 °C, 300 MPa; (e) NDT + 60 °C, 400 MPa.
- The stress that matters is the total one, applied **plus residual**. A welded vessel is loaded to a membrane stress of 100 MPa, a third of S_y . Why does it still have to be pressurised above FTE?
- A vessel is brought to pressure at no less than 4 °C (40 °F), and its total stress reaches S_y . What is the highest NDT the steel may have? What is the most stress the vessel may carry below that NDT?

4. The steel is bought with $NDT = -30\text{ }^{\circ}\text{C}$ ($-22\text{ }^{\circ}\text{F}$). After years of service, surveillance specimens show an NDT shift of $50\text{ }^{\circ}\text{C}$ ($90\text{ }^{\circ}\text{F}$).
 - (a) Where are NDT, FTE and FTP now?
 - (b) May the vessel still be brought to full pressure at $4\text{ }^{\circ}\text{C}$?
 - (c) What is the lowest temperature at which it may?
5. Why does irradiation move NDT to the right? Answer with the Davidenkov construction: the yield stress against the cleavage fracture stress.
6. (*Jawad and Farr, 2019, Example 4.2.*) A low-carbon steel with $NDT = 15\text{ }^{\circ}\text{F}$ ($-9\text{ }^{\circ}\text{C}$) is used in a pressure vessel. What is the minimum safe operating temperature? What if stress concentrations take the stress beyond yield in some areas?
7. (*Jawad and Farr, 2019, Example 4.3.*) A low-carbon steel vessel with $NDT = -20\text{ }^{\circ}\text{F}$ ($-29\text{ }^{\circ}\text{C}$) is to start up at $0\text{ }^{\circ}\text{F}$ ($-18\text{ }^{\circ}\text{C}$), at a stress of half the yield strength. Is the start-up temperature safe?

Solution

1. Reading the diagram

The crack-arrest curve D runs along the lower fracture propagation stress, 34-55 MPa, up to NDT, then rises through $\frac{1}{2}S_y = 150\text{ MPa}$ at $NDT + 17\text{ }^{\circ}\text{C}$, $S_y = 300\text{ MPa}$ at $FTE = NDT + 33\text{ }^{\circ}\text{C}$ and $S_u = 480\text{ MPa}$ at $FTP = NDT + 67\text{ }^{\circ}\text{C}$.

case	where	verdict
(a) $NDT - 10\text{ }^{\circ}\text{C}$, 30 MPa	under the lower fracture propagation stress	arrested: below 34 MPa no crack runs, at any temperature
(b) $NDT - 10\text{ }^{\circ}\text{C}$, 150 MPa	above it, left of curve D	propagates
(c) $NDT + 45\text{ }^{\circ}\text{C}$, 250 MPa	right of FTE, below S_y	arrested: past FTE no crack runs under elastic stress
(d) $NDT + 20\text{ }^{\circ}\text{C}$, 300 MPa	at yield, left of FTE	propagates: FTE is $NDT + 33\text{ }^{\circ}\text{C}$
(e) $NDT + 60\text{ }^{\circ}\text{C}$, 400 MPa	right of FTP, above yield	no brittle fracture: past FTP failure is by shear

Cases (c) and (d) carry nearly the same stress and differ only in temperature. That is the point of the diagram: whether a stress is safe depends on how far above NDT the steel is.

2. Residual stress

A weld leaves residual stresses that stress relief does not remove entirely. The amount left can conservatively be taken as of the order of the yield strength (Harvey, 1974, §5.22). The total, applied plus residual, then reaches S_y even though the applied load is only a third of it. A crack can run at S_y anywhere left of FTE, so the vessel has to be above FTE before it is pressurised.

3. Choosing a steel

At S_y the vessel has to be at or above FTE when it is pressurised:

$$NDT + 33\text{ }^{\circ}\text{C} \leq 4\text{ }^{\circ}\text{C} \Rightarrow NDT \leq -29\text{ }^{\circ}\text{C} (-20\text{ }^{\circ}\text{F})$$

In $^{\circ}\text{F}$ the data are round: $40 - 60 = -20\text{ }^{\circ}\text{F} = -28.9\text{ }^{\circ}\text{C}$.

Below NDT the total stress must stay under the lower fracture propagation stress, **34-55 MPa (5-8 ksi)**: Chattopadhyay (2005) takes the low end, 34.5 MPa. On a cold vessel that is a small fraction of the operating stress (Weisman, 1977, §10.4).

4. A vessel ages

- (a) Every temperature on the diagram is counted from NDT, so the whole diagram moves by the shift:

	as bought	after the shift
NDT	-30 °C (-22 °F)	20 °C (68 °F)
FTE = NDT + 33 °C	3 °C (38 °F)	53 °C (128 °F)
FTP = NDT + 67 °C	37 °C (98 °F)	87 °C (188 °F)

- (b) **No.** At 4 °C the steel is now 16 °C *below* its NDT: only the lower fracture propagation stress, 34-55 MPa, is allowed. When new, 4 °C was just above FTE and the vessel could be loaded to yield.
- (c) Full pressure needs $T \geq \text{FTE}$: **53 °C (128 °F)**. The vessel has to be heated before it is pressurised, and the minimum pressurisation temperature has risen by the whole shift, 50 °C.

5. Why irradiation moves NDT

Neutrons fill the lattice with defects that obstruct dislocations. The yield stress rises at every temperature, while the cleavage fracture stress hardly changes. The temperature where the two cross, below which the steel cleaves before it can yield, moves up. So does NDT, and with it the whole diagram.

6. Minimum safe operating temperature (Jawad and Farr, Example 4.2)

No stress level is given, so the stress is assumed at yield. At S_y the crack-arrest curve is crossed at FTE:

$$T_{\min} = \text{NDT} + 60\text{ °F} = 15 + 60 = 75\text{ °F} (24\text{ °C})$$

If stress concentrations take the stress beyond yield in some areas, the conservative choice is FTP:

$$T_{\min} = \text{NDT} + 120\text{ °F} = 135\text{ °F} (57\text{ °C})$$

In °C: NDT = -9.4 °C, plus 33.3 °C and 66.7 °C, gives 23.9 °C and 57.2 °C.

7. A cold start-up (Jawad and Farr, Example 4.3)

At half the yield strength the crack-arrest curve gives NDT + 30 °F:

$$T_{\min} = -20 + 30 = 10\text{ °F} (-12\text{ °C})$$

The start-up at 0 °F (-18 °C) is below it, so it is **unsafe**. Either the stress is reduced during start-up, or a steel with a lower NDT is chosen.

A limit of the diagram that matters for a reactor vessel. Jawad and Farr (p. 39) give two conditions for using these steps: the steel has to be a low-carbon steel, and the section less than 2 in (51 mm) thick. Above 6 in (152 mm) it has been proposed to take FTE = NDT + 120 °F (67 °C) instead of NDT + 60 °F, and FTP = NDT + 210 °F (117 °C) instead of NDT + 120 °F. A reactor vessel wall is well over 6 in thick, so for it the diagram as drawn is on the unconservative side.